

Introduction to Future Contracts

- Definition: A standardized legal agreement to buy or sell an asset at a predetermined price at a specified time in the future.
- Purpose:
 - Hedging: To mitigate price risk (e.g., a farmer selling futures to lock in a price for their crop).
 - Speculation: To profit from anticipated price movements.
 - Arbitrage: To profit from price discrepancies in different markets.
- Key Characteristics:
 - Standardized: Contract size, quality, delivery date, and location are specified by the exchange.
 - Exchange-traded: Traded on organized exchanges (e.g., NSE, BSE in India).
 - Marked-to-market: Profits/losses are settled daily.
 - Leverage: Small margin deposit controls a large contract value.
 - Obligation: Both buyer (long) and seller (short) are obligated to fulfill the contract.

Future Contracts Terminology

- Underlying Asset: The asset on which the future contract is based (e.g., shares, commodities, currency, index).
- Spot Price: The current market price of the underlying asset for immediate delivery.
- Future Price: The price agreed upon today for future delivery of the underlying asset.
- Expiration Date (Maturity Date): The date on which the contract ceases to exist.
- Contract Size: The amount of the underlying asset covered by one contract.
- Tick Size: The minimum price fluctuation for a future contract.
- Margin:
 - Initial Margin: Deposit required to open a futures position.
 - Maintenance Margin: Minimum margin balance that must be maintained.
 - Margin Call: A demand for additional funds if the margin account falls below maintenance margin.
- Long Position: Buying a futures contract, expecting the price to rise.
- Short Position: Selling a futures contract, expecting the price to fall.

Mechanism of Trading Futures

1. Opening a Position:
 - Buy (Long): Trader places an order to buy a specific futures contract.
 - Sell (Short): Trader places an order to sell a specific futures contract.
 - Requires depositing initial margin with the broker.
2. Clearing House: Acts as an intermediary, guaranteeing performance of both buyer and seller. Eliminates counterparty risk.
3. Mark-to-Market:
 - At the end of each trading day, the contract is revalued to its current market price.
 - Profits are credited to the margin account, and losses are debited.
 - If the margin account falls below maintenance margin, a margin call is issued.
4. Closing a Position:
 - Offsetting Trade: Most common method. A long position is closed by selling an identical contract, and a short position is closed by buying an identical contract.
 - Delivery (rare for financial futures): Physical delivery of the underlying asset. More common in commodity futures.
 - Cash Settlement: The difference between the future price and the spot price at expiration is settled in cash. Common for index futures.

Pricing of Future Contracts

- Cost of Carry Model: The theoretical future price (F_0) is determined by the spot price (S_0) and the cost of holding the asset until maturity.
$$F_0 = S_0 * (1 + r - d)^T$$
Where:
 - F_0 = Future Price
 - S_0 = Spot Price
 - r = Risk-free interest rate (annualized)
 - d = Dividend yield (for stocks/indices) or storage costs (for commodities) (annualized)
 - T = Time to expiration (in years)
- For Non-Dividend Paying Stocks/Commodities (with storage costs):
$$F_0 = S_0 * e^{(r+c)T}$$
Where c = storage cost as a percentage of asset value.
- For Stock Index Futures:
$$F_0 = S_0 * e^{(r-q)T}$$
Where q = dividend yield on the index.
- For Currency Futures (Interest Rate Parity):
$$F_0 = S_0 * \frac{(1 + r_{domestic})^T}{(1 + r_{foreign})^T}$$

Where $r_{domestic}$ and $r_{foreign}$ are domestic and foreign risk-free rates, respectively.

- Contango vs. Backwardation:
 - Contango: Future price > Spot price ($F_0 > S_0$). Occurs when cost of carry is positive.
 - Backwardation: Future price < Spot price ($F_0 < S_0$). Occurs when cost of carry is negative (e.g., high convenience yield for commodities).
- Factors Affecting Future Prices:
 - Spot price of the underlying asset
 - Time to expiration
 - Risk-free interest rate
 - Dividends (for stocks/indices) or storage costs (for commodities)
 - Supply and demand dynamics
 - Market sentiment and volatility

Hedging Strategies with Futures

- Long Hedge: Buying futures contracts to protect against a future increase in the price of an asset (e.g., a manufacturer needing raw materials later).
- Short Hedge: Selling futures contracts to protect against a future decrease in the price of an asset (e.g., a farmer selling crops that will be harvested later).
- Basis Risk: The risk that the relationship between the spot price and the futures price (the basis) changes unexpectedly.
 - Basis = Spot Price - Futures Price
 - Hedging reduces price risk but introduces basis risk.